# A Task-Balanced Multiscale Adaptive Fusion Network for Object Detection in Remote Sensing Images

Tao Gao, *Member, IEEE*, Zixiang Liu, Jing Zhang, Guiping Wu, and Ting Chen

***Abstract*— Object detection is essential in the interpretation of remote sensing images (RSIs). However, the blurred background and objects with vast variances are identified as the two main challenges of the task. We propose a novel detector adapted to a complicated background and multiscale objects, namely, task-balanced multiscale adaptive fusion network (TMAFNet), targeting directly on the above two challenges. First, a depth separable global context (DSGC) module is constructed to understand contextual relations among pixels from a global perspective, which is extraordinarily necessary to distinguish objects from the environment. Most importantly, DSGC reduces the computational cost by decoupling the acquisition of global information into single-channel global interaction and multichannel single-point interaction. Second, in order to eliminate disturbance and enhance the representation ability of objects, hidden recursive feature pyramid network (HRFPN) is explored, which encodes the information of difference before and after using the multiscale fusion. HRFPN is proven to enhance the target features by reducing the background noise. Third, a semicoupling task-balanced (SCTB) head is presented to guarantee the consistency of detection. We have conducted comprehensive experiments on several publicly available datasets, and the results illustrate that our modules improve adaptability and robustness of the network, leading to a new state-of-the-art.**

***Index Terms*— Contextual relationship modeling, hidden recursive feature pyramid network (HRFPN), object detection, remote sensing, task-balanced multiscale.**

## I. Introduction

OBJECT detection is a fundamental task in the analysis of remote sensing images (RSIs), which is dedicated to implementing two tasks: localization and multiclass classification. Given the vast amount of RSIs data, it is feasible to design accurate and efficient detection models for RSIs. As a convenient way for RSIs interpretation, object detection of RSIs plays an indispensable role in wide applications, such as intelligent monitoring, traffic control, emergency rescue, military field, and more [1], [2], [3], [4], [5].

Manuscript received 12 March 2023; revised 21 May 2023 and 19 June 2023; accepted 21 June 2023. Date of publication 27 June 2023; date of current version 6 July 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2019YFE0108300; in part by the National Natural Science Foundation of China under Grant 52172379, Grant 62001058, and Grant U1864204; and in part by the Special Funds for Fundamental Research Funds of the Central Universities of Chang'an University under Grant 300102242901. *(Corresponding authors: Tao Gao; Ting Chen.)*

Tao Gao, Zixiang Liu, Guiping Wu, and Ting Chen are with the Department of Information Engineering, Chang'an University, Xi'an 710064, China (e-mail: tgaochd@126.com; tchenchd@126.com).

Jing Zhang is with the School of Computing, The Australian National University, Canberra, ACT 2600, Australia.

Digital Object Identifier 10.1109/TGRS.2023.3289878

Over the past decades, benefiting from the rapid development of deep learning technology, convolutional neural networks (CNNs) [6] exhibit powerful extraction capabilities and attract the attention of academic experts. Therefore, object detection based on deep learning has gradually replaced traditional detection algorithms and achieved satisfactory results. In general, current detection algorithms are mainly divided into three categories as follows.

1) Faster R-CNN [7] is the representative work for "two-stage" detection, which implements detection by region extraction and classification of bounding boxes. Despite their excellent accuracy, their performance is restricted by an enormous amount of calculation and time overhead.
2) YOLO, SSD, and Retinanet [8], [9], [10], [11], [12], [13] regard detection as a regression problem for single network. They divide the image into different grids, using the output of each grid to predict positioning boxes and categories of targets. Thus, they acquire relatively balanced results between speed and accuracy.
3) Since the above networks basically employ default boxes, which not only bring mass calculation, but also need to set more hyperparameters. To strengthen the robustness of detectors, fully convolutional one-stage object detection (FCOS), Centernet, Cornernet, and other anchor-free methods [14], [15], [16], [17] emerged. Instead of setting up default boxes in advance, they directly predict the position, height, and width of the target, with better adaptability.

Recently, feature pyramid network (FPN) and some improved algorithms [18], [19], [20] based on it innovatively integrate information of multilevel features, which boost the detection accuracy of multiscale targets. Attention mechanisms such as efficient channel attention module (ECA) [21], convolutional block attention module (CBAM) [22], and spatial attention module (SAM) [23] focus on the information that is more critical to the current task, which greatly facilitates the adaptability of the network to different environments. In addition, with the advent of self-attention [24] and windows multihead self-attention (W-MSA) [25], the network is easier to mine the global context, and the relationship between pixels also becomes clearer.

Although the previous methods achieve great success for natural images, we argue that most of them perform poorly in

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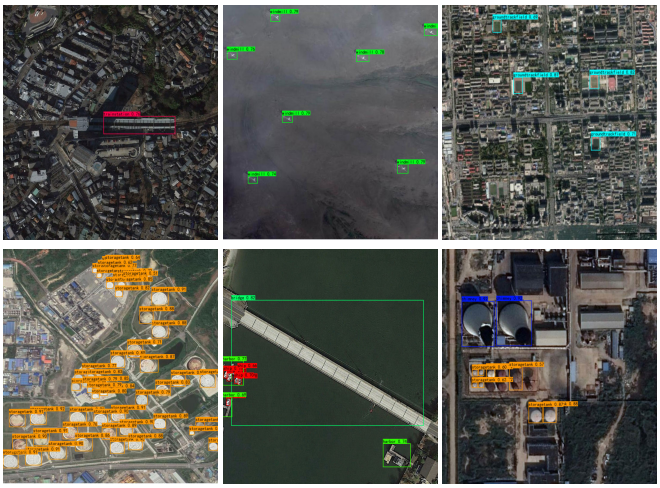


Fig. 1. Remote sensing object detection is characterized by the broad coverage range, which introduces complex environmental interferences such as geographic and architectural textures, light and shadow, and cloud obscuration, as well as the variable shape of multiscale objects, including shape distortion and extreme aspect ratios.

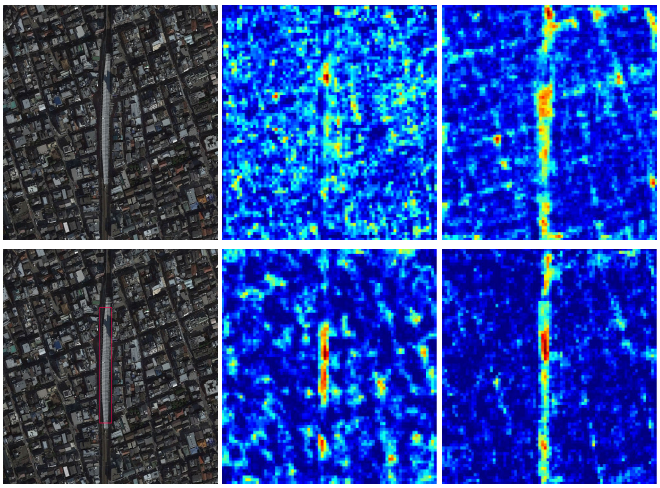



Fig. 2. Visualization of detection results. (a) Original image and visualization of detection result. (b) Heat maps of decoupled classification and location branches. (c) Heat maps of our classification and location.

the detection of RSIs due to the following obstacles illustrated in Fig. 1.

1) Remote sensing imagery captures vast geographical areas with intricate environmental factors, including terrain and building textures, image blurring, shadow variations, and cloud occlusions, all of which may contribute to targets becoming submerged in complex backgrounds.
2) The significant scale and shape variations of objects, along with their nonuniform distribution, pose a challenge for object detection in remote sensing imagery. In particular, targets with extreme aspect ratios may exhibit irregular distortions caused by variations in imaging altitude and angle.

Considering the above-mentioned characteristics of RSIs, many improved algorithms have been introduced. For interference from a complex environment, Tian et al. [26] presented a relation-augmented embedded graph attention network (EGAT), which fully exploited underlying spatial and semantic relations among objects for improving detection performance. Yuan and Zhang [27] optimized the positioning accuracy by providing more accurate regions of interest to train better classifiers. Hong et al. [28] proposed miniGCN, which not only improved the ability to model the relationships between samples, but also effectively achieved the purpose of training large-scale graph convolutional network (GCN) with small batches. More importantly, it can provide more diverse and discriminative feature representations for the current task without retraining the network and improving classification performance. To overcome the limitation of information diversity in complex scenes, Hong et al. [29] developed a general multimodal deep learning (MDL) framework. This framework is not only applicable to pixel classification tasks but also effective for spatial information modeling of CNNs. In addition, Liu et al. [30] constructed a global perception network (GPNet) to preserve the relationships of the entire dataset and capture the dependencies between the space and channel. For objects with vaster scale variance, Li et al. [31] designed a novel structure-guided feature transform hybrid residual (SGFTHR) network, which conquered the low performance of detection of objects at different scales. Wu et al. [32] proposed a novel object detection framework, which exhibits better robustness to the deformation of various objects by integrating diverse channel feature extraction, feature learning, fast image pyramid matching, and boosting strategy. Zhang et al. [33] proposed a new synthetic fusion pyramid network (SFPN) architecture that creates various synthetic layers among layers of the original FPN. Wu et al. [34] developed an efficient "U-Net in U-Net" framework that treats detection as a segmentation problem and uses multiscale learning of U-Net to achieve superior multiscale object detection in remote sensing with no interference. This approach has a strong generalization ability and is highly robust.

Although these approaches have achieved outstanding performance, there are some deficiencies to be alleviated. First, despite previous methods that have explored contextual information, they seldom capture both global and local description with a balance between effectiveness and efficiency. Second, many feature enhancement networks present a variety of complex paths to efficaciously fuse multiscale information. However, such indiscriminately direct integration of multiple features not only causes redundancy in calculations but also introduces new aliasing noises. In addition as shown in Fig. 2, previous methods usually ignored that it is extremely necessary to provide different representation for two tasks and maintain the consistency between them.

Taking the above matters into consideration, we present a task-balanced multiscale adaptive fusion network (TMAFNet). First of all, a depth separable global context (DSGC) module is proposed to capture global and local contextual information without much computational costs, which allows our network to distinguish the classes and locations of objects with the help of environmental information. Second, a hidden recursive

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FPN (HRFPN) is created to selectively enhance the expression of multiscale targets. HRFPN only performs one fusion of adjacent feature maps, which will not bring information obscuration of different levels. And by optimizing the change of information after fusion and reinjecting it into original features, the target information is further strengthened and the background information is suppressed. Third, the semi-coupling task-balanced (SCTB) head is explored to provide the required quality presentation for the classification and positioning through the partial channel fusion state module and the multidirectional distance attention module, respectively. At the same time, the consistent alignment of the two tasks is achieved.

Our main contributions are summarized as follows.

1) We propose the DSGC module to attain a more global description of relationships of environments and detect targets. Compared with other methods for global information mining, DSGC module decreases computational cost by decoupling the acquisition of global context into single-channel global interaction and multichannel single-point interaction.
2) HRFPN is exploited to enhance representational capacity of multiscale targets and filter interference from complex environment by focusing on the amount of variation in information and injecting the optimized information into the fused features.
3) According to different needs of classification and localization, SCTB detection head is designed to provide information with different focuses for two sub-tasks. Furthermore, in order to achieve more consistent results in two tasks, the SCTB head also offers a better balance between learning task-interactive and task-specific features.

## II. Related Work

### A. Contextual Information Exploitation

The exploration of contextual information is necessary to comprehend the relationship between complex environments and goals. Many ground-based methods proved that exploiting the contextual relationship is beneficial to work out the problems of insufficient representation of object appearance and alleviate environmental interference. Therefore, many effective modules are introduced to extract local and global contextual information. SAM [23] was proposed to exploit local contextual expression while transformer [25] was created to capture global relation between pixels. Wu et al. [34] proposed an interactive-cross attention module that encodes local contextual information between low-level details and high-level semantic features through the outputs of two subtasks, cross-channel attention, and interactive-cross spatial attention. Especially for RSIs, there are many disturbing factors such as lighting, shadows, and similar objects in the overhead imaging process, so many networks proposed new solutions. For example, full-scale object detection network (FSoD-Net) [35] established three branches to capture various contextual information for multiscale object detection. Sun et al. [36] proposed a category-position (CP) module to generate guidance vectors from the classification branch, aiming to enhance the localization capability of the network in complex scenes. They also redesigned the classification and regression methods to mitigate the adverse effects of ambiguous regions. Lin et al. [37] addressed the problem of inaccurate sea-land masks in ship detection by using the squeeze and excitation mechanism to obtain richer contextual information, resulting in further improvement of detection accuracy. An enhanced effective channel attention (EECA) and a context enhancement module (CEM) are designed by adaptive balanced network (ABNet) [38] to exploit abundant semantic and detailed information. Liu et al. [30] also explored GPNet, which captured the dependencies between the space and channel and combined the attention information to enhance the network model. Therefore, DSGC is proposed to promote detection accuracy by generating local and global context without abundant calculation.

### B. Multiscale Information Fusion

Early YOLO and RCNN [39] both output the last layer of features for detection, which received serious limitations in multiscale object detection. FPN innovatively incorporates and outputs information at different scales through a top-down path, which greatly improves the detection performance of the network for multiscale targets. Since then, many improved structures based on FPN have gradually appeared [40], [41]. For example, path aggregation network (PANet) [19], bidirectional FPN (BiFPN) [42], one-shot PANet architecture search (OPANAS) et al. constructed various heterogeneous information paths to aggregate multiscale information efficiently. Recently, many FPN-based improved algorithms have achieved remarkable achievement in view of the characteristics of RSIs. For example, Liu et al. [43] proposed scene-enhanced FPN (SE-FPN) to capture contextual features of objects with the help of the scene features. Sun et al. [44] proposed a novel bidirectional feature fusion module (Bi-DFFM) that can effectively aggregate multiscale features by bidirectional (top-down and bottom-up) information interaction, which is helpful for detecting ships of various scales. Kang et al. [45] designed a contextual region-based CNN with multilayer fusion, which fuses the deep semantic and shallow high-resolution features by using low-resolution feature maps from a single layer for proposal generation in a region proposal network (RPN). In order to combine multiscale features adaptively in different channels and spatial positions, Liu et al. [38] introduced adaptive FPN (AFPN) to obtain more discriminative features. Wang et al. [46] proposed a feature-reflowing pyramid (FRP) structure to generate high-quality feature presentation for each scale by fusing fine-grained features from the adjacent lower level. In this article, we propose HRFPN, which effectively extinguishes noises and highlights targets without causing calculation duplication and information redundancy.

### C. Task-Balanced Detection

In general, object detection is composed of two tasks: localization and classification. Bounding-box regression requires abundant spatial context information, while classification needs rich semantic expression. Therefore, as depicted in Fig. 1, differences in learning mechanisms inevitably cause

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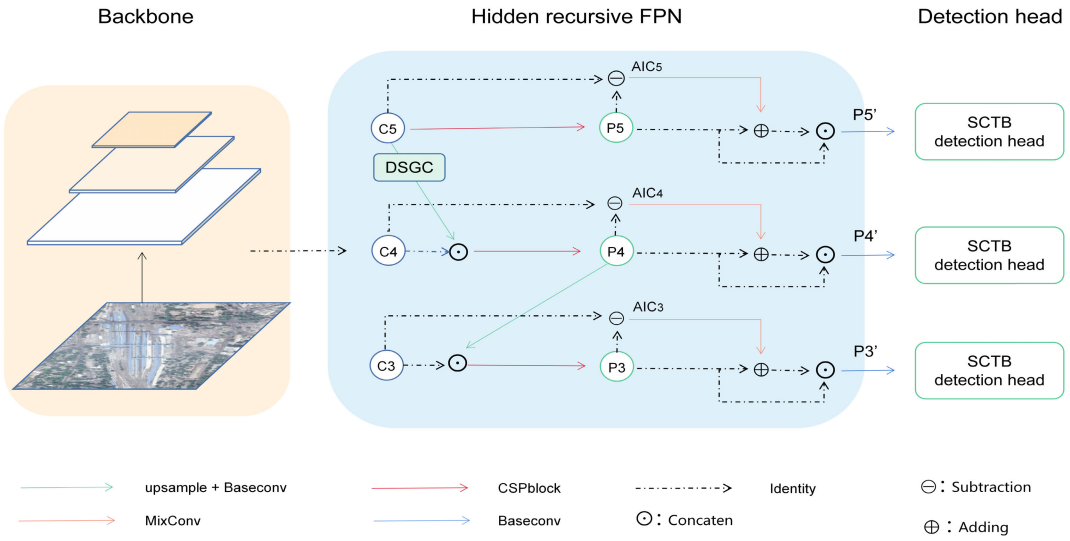



Fig. 3. Overview of multiscale adaptive fusion anchor-free network for RSIs. "DSGC" is the depth separable global context module, "FPN" is the feature pyramid network, and "SCTB" is the semicoupling task-balanced head.

imbalance and misalignment between classification score and localization. Furthermore, vaster variance of objects and interference from the background might aggravate the problem. With the intention of remedying these matters, Feng et al. [47] developed a novel Task-aligned Head, which offers a better balance between learning task-interactive and task-specific features. Zhang et al. [48] proposed a task-collaborated detector (TCD) that utilizes the TCA component to identify accurate and consistent positive anchor boxes. Additionally, the task-collaborated head (TCH) in TCD enables flexible adjustment of the spatial feature distribution for classification and localization tasks through the learning of joint features from the aggregation layer. Tang et al. [49] designed a task-decoupled feature distillation (TFD) to generate harmonious features by flexibly balancing the contributions of classification and regression tasks. Furthermore, a feature alignment module (FAM) was created by Han et al. [50] to generate high-quality anchors with an anchor refinement network and adaptively align the convolutional features with a novel alignment convolution. Feature aligned single-shot detection (ASSD) [51] devised context-based FAM (CFAM), which adaptively adjusted the sampling points of the convolution kernel and collects the context information according to the aligned core anchor. Different from the strategies described above, SCTB not only provides both tasks with the information they need individually

## III. Proposed Method

### A. Framework Overview

As illustrated in Fig. 3, our network is composed of a ResNet50 [52] backbone, an HRFPN, an SCTB head, and an anchor-free detection framework. First of all, ResNet50 is adopted to implement feature extraction on the original image effectively and generate multilevel feature expressions. Second, DSGC mines global and local contextual representation and completes the extraction of global relation between targets and backgrounds. It also can utilize image-level labels to capture information relationships between global and local areas. To be specific, it decouples acquisition of global information into global spatial information and global channel information, significantly reducing computational overhead. After that, HRFPN propagates global information to multiscale features through a top-down path. By optimizing changes in features after fusion of multiscale expressions and repeatedly injecting them into the original feature maps, HRFPN strives to highlight representation of targets and suppress background noises. Finally, SCTB head not only provides distinct descriptions for classification and localization, but also better coordinates the connection between them. The details of DSGC module, HRFPN network, and SCTB head are depicted as follows.

### B. Depth Separable Global Context Module

As the network depth increases, the detailed information of targets from complex environment is gradually lost. Some studies have shown the importance of using context for object detection [53], [54], [55], and ABNet [38] believes that extracting contextual information for objects in complex backgrounds of RSIs is even more important. DSGC, as a global context extraction module, it differs significantly from traditional nonlocal methods such as self-attention and W-MSA. Instead of unfolding 3-D features into 2-D and performing matrix multiplication to achieve direct interaction between global information, DSGC extracts global spatial information within a single channel first and then propagates information among multiple channels to the entire feature map through point convolution. By decoupling spatial and channel information within 3-D features, DSGC can reduce the computational cost by an order of magnitude compared to traditional global context extraction methods. At the same time, $C_5$ serves as the deepest output of the backbone feature network, providing the highest-level feature information. And each element in its spatial dimension corresponds to the region with the largest

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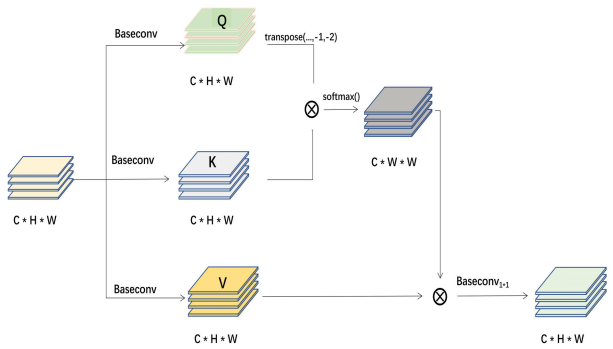



Fig. 4. Illustrate of DSGC module.

span in the original image. This characteristic makes it an ideal candidate for applying DSGC to capture global contextual information more effectively. Second, $C_5$, as the top level of the FPN, can propagate the globally extracted feature information through a top-down propagation path and supplement the multiscale features of $C_4$ and $C_3$. To achieve trade-off between time and space complexity, as shown in Fig. 4, a plug-and-play DSGC module is created to decouple the acquisition of global context of $C_5$ into two steps. First, acquisition of the global spatial context for each channel: First of all, The input feature $X_{\text{in}}$ is adopted to generate three learnable context features $Q, K, V \in \mathbb{R}^{C\times H\times W}$ by offering three parallel Baseconv branches. Specially, Baseconv$_{i\times i}$ is combination of convolution (kernel size $= i$), batch normalization and sigmoid weighted linear unit (SiLU) activation function. Then, a matrix multiplication operation is performed between $K$ and the transpose of $Q$ to obtain the correlation information of the global space within each channel. Furthermore, correlation information is scaled and activated by the sigmoid function to transmit the weight of different elements in space. Finally, global spatial context information is extracted by performing a matrix multiplication between the weight information and $V$. This step efficiently carries out spatial information interaction with low computation. Second, the interaction of information among multiple channels: The previous step is independent among different channels, so the convolution operation of $1 \times 1$ is utilized to interact with the same location of the multiple channels. All elements participate in the extraction and propagation of information through the decoupling and coupling operations. As shown in the following equation, DSGC reduces the computational amount of matrix operation to $1/C$ of the previous method in which all elements participate directly in the calculation

$$\frac{\text{Our}}{\text{PRE}} = \frac{(H*W)^2 * C}{(H*W*C)^2} = \frac{1}{C} \tag{1}$$

where $C$, $H$, and $W$ represent the number of channels, width, and height of input features, respectively. And the processing workflow can be summarized as shown in the following equation:

$$\begin{aligned} Q, K, V &= \text{Baseconv}(X_{\text{in}}) \\ \text{New}V &= \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}} \cdot V\right) \\ X_{\text{out}} &= \text{Baseconv}_{1\times 1}(\text{New}V). \end{aligned} \tag{2}$$

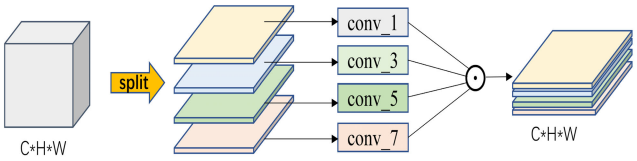



Fig. 5. Architecture of mixed convolution. Conv_$i$ denotes 2-D convolution of kernel size $i$. Split refers to dividing the channels into four equal parts.

### C. Hidden Recursive Feature Pyramid Network

Considering complicated environments and shape variations of targets in RSIs, more adequate and balanced feature representation is essential for multiscale object detection [56]. Through the fusion of features from different hierarchical levels, the network is capable of strengthening its capability in representing target features, while also mitigating the effect of noise interference to a certain extent [43], [57], [58]. Most existing multiscale feature fusion networks prioritize aggregating multiscale feature information effectively by designing multiple complex fusion paths [20], [41], [46]. Compared with other enhancement strategies, the proposed HRFPN tends to assign more importance to the change of information and repeatedly reinforce representation of targets without requiring massive calculations, which allows us to further amplify the information in the target region. As is illustrated in Fig. 3, $C_3$, $C_4$, and $C_5$ are multiscale feature levels from ResNet50 backbone network. First, FPN completes combination and enhancement with neighboring features through a top-bottom path. At the same time, CSPblock module obtains $P_3$, $P_4$, and $P_5$ by selectively incorporating effective targets information and filtering strong background noises in the integration of multiscale features.

Compared with $C_i$ and $P_i$ not only has richer semantic and spatial detailed information of targets due to the assimilation of supplementary information from adjacent features but also alleviates environmental interference. Therefore, by eliminating $C_i$ from $P_i$, the amount of information changed (AIC$_i$, $i \in 3, 4, 5$) after multiscale fusion is acquired, including the varying degrees of enhancement of target information and suppression of environmental noises. Later, the mixed convolution operation is utilized to further screen and correct AIC$_i$. As shown in Fig. 5, the mixed convolution operation of different kernel sizes (1, 3, 5, 7) is conducted for different depths, and the influence degree of details and semantics is strengthened through multiple receptive fields of various sizes. By optimizing the change amount of information through mixed convolution and feeding it back to the fused feature layer, multiple feature fusions and dynamic adjustment are achieved. Specifically, it achieves subsequent information enhancement by reinjecting optimized feature changes, rather than requiring the involvement of other scale features. The process is summarized as follows:

$$\begin{aligned} \text{AIC}_i &= P_i - C_i, \quad i = \{3, 4, 5\} \\ P_i' &= (\text{Mixconv}(\text{AIC}_i) + P_i) \odot P_i \end{aligned} \tag{3}$$

where $\odot$ represents stack operation and Mixconv denotes mixed convolution operation.

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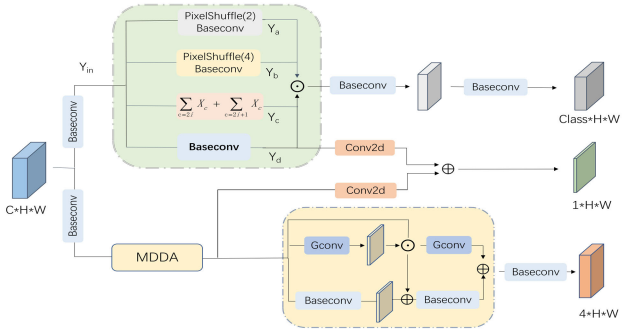



Fig. 6. Illustrate of SCTB head. MDDA denotes multidirectional distance attention and Class represents the number of categories.

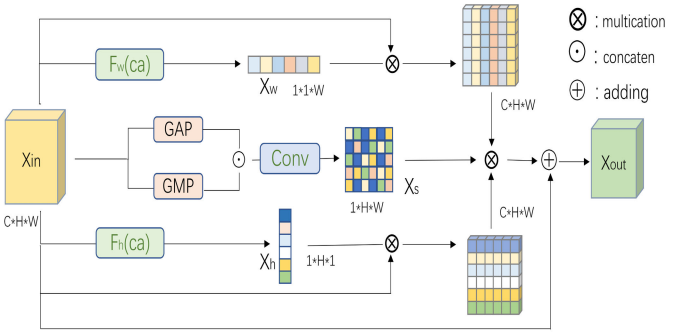



Fig. 7. Architecture of multidirectional distance attention. $F_w$(ca) and $F_h$(ca) denote the transpose of channel attention in different directions.

Our proposed "hidden recursive" strategy focuses on the difference between the information before and after the multiscale feature fusion process. By optimizing the change amount of information and reinjecting it into the fused feature layer, a second-level multiscale fusion operation is obtained and positive feedback is dynamically provided to address possible information changes. This implicit recursive operation is conceptually similar to the recursive feature pyramid (RFP) but differs in its implicit nature. Specifically, it achieves subsequent information enhancement by reinjecting optimized feature changes, rather than requiring the involvement of other scale features. This mechanism is referred to as "hidden." This module is highly extended and can be adopted for multiple iterations.

### D. Semicoupling Task-Balanced Head

Although decoupled heads have long been one of the standard paradigms for single-stage detection, recent research studies [47], [49], [50] have demonstrated that adopting two parallel branches to implement classification and localization tasks may lead to a certain degree of spatial inconsistency and imperfect balance between the two tasks. The extreme aspect ratio of targets and diverse backgrounds may aggravate the imbalance of two tasks. Considering the above-mentioned shortcomings, SCTB detection head is created to alleviate previous issues. As demonstrated in Fig. 6, two parallel $\text{Baseconv}_{3\times3}$ branches are adopted to guide the input features to generate descriptions suitable for classification and localization, respectively. For classification branch, multiple depth fusion module is explored to produce abundant semantic expression, which aggregates information about the multichannel view through four branches. To be specific, $Y_{\text{in}}$ first performs pixelshuffle operations at different scales for two of the branches to conduct the coupling between channels and space, and then downsample through the convolution operation to obtain richer semantic information. The other two branches, one refines the features using a convolution of $3 \times 3$, and the other one directly adds up to the parity of the adjacent channels to form the aggregation of analogical information. Finally, the four outputs are combined, and their information is fused through the $\text{Baseconv}_{3\times3}$ operation to obtain the description for classification. The entire process can be summarized as followed:

$$
\begin{aligned}
Y_a &= \text{Baseconv}_{3\times3}(\text{PixShuffle}_2(Y_{\text{in}})) \\
Y_b &= \text{Baseconv}_{3\times3}(\text{PixShuffle}_4(Y_{\text{in}})) \\
Y_c &= \sum_{c=2i} Y_{\text{in}}c \oplus \sum_{c=2i+1} Y_{\text{in}}c \\
Y_d &= \text{Baseconv}_{3\times3}(Y_{\text{in}})
\end{aligned} \tag{4}
$$

where Pixshuffle_$i$ indicates $i$ times of upsampling operation. For the localization branch, in order to better ignore vaster shape variances of objects and disturbances from complex environments, multidirectional distance attention is exploited to obtain comprehensive spatial details and distance expression. As shown in Fig. 7, the input expression $X_{\text{in}} \in \mathbb{R}^{C\times H\times W}$ is pooled in both horizontal and vertical directions to perform perceptual compression for obtaining global horizontal distance information $X_w \in \mathbb{R}^{1\times1\times W}$ and global vertical distance information $X_h \in \mathbb{R}^{1\times H\times1}$. Meanwhile, spatial attention is applied to $X_{\text{in}}$ to extract spatial contextual information $X_s \in \mathbb{R}^{1\times H\times W}$. This process is summarized as follows:

$$
\begin{aligned}
X_{\text{in}}^h &= \text{Permute}(X_{\text{in}}, H, C, W) \\
X_{\text{in}}^w &= \text{Permute}(X_{\text{in}}, W, H, C) \\
\text{X}_{\text{s}} &= \sigma\text{Conv}_{3\times3}[\text{GAP}(X_{\text{in}}); \text{GMP}(X_{\text{in}})] \\
X_h &= \sigma\, P\left(\text{Conv}_{3\times3}\left(\max\left(X_{\text{in}}^h\right)\right) + \text{Conv}_{3\times3}\left(\text{avg}\left(X_{\text{in}}^h\right)\right)\right) \\
X_w &= \sigma\, P\left(\text{Conv}_{3\times3}\left(\max\left(X_{\text{in}}^w\right)\right) + \text{Conv}_{3\times3}\left(\text{avg}\left(X_{\text{in}}^w\right)\right)\right)
\end{aligned} \tag{5}
$$

where GAP and GMP denote global average pooling and global max pooling, respectively. Max and avg refer max-pooling and average pooling at the spatial level. $\sigma$ represents sigmoid function. Permute and $P$ denote transposing of the matrix. $\text{Conv}_{3\times3}$ denotes $3 \times 3$ convolution operation. After that, we obtain the feature $X_{\text{out}}$, which is sensitive to both spatial context and multidirectional distance information, through the following formula:

$$
X_{\text{out}} = \text{ReLu}(X_{\text{in}} \oplus (X_{\text{in}} \otimes X_h) \otimes X_s \otimes (X_w \otimes \text{X}_{\text{in}})) \tag{6}
$$

where $\otimes$ denotes elementwise multiplication and $\oplus$ refers element-by-element addition. After that, while carrying out ordinary convolution, group convolution residual module branch is introduced to strengthen the extraction of local channel information. By increasing the number of groups, local information is easier integrated with global situation. Finally, Local and global channel information are combined and fully integrated by Baseconv operations to comprehensive description for positioning. The confidence score in object detection represents the evaluation of the accuracy of the

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model's predictions, which takes into account not only the probability of the object category but also the accuracy of the object position and size. The designed SCTB detection head utilizes confidence scores as a bridge for dynamic communication between the two subtasks. Furthermore, multiple depth fusion module enhances the semantic information required for the classification branch by coupling different levels of spatial and channel information. The multidirectional attention models the local contextual information, global horizontal, and vertical distance information separately in the spatial dimension to highlight the details relevant to localization in the spatial domain. Thus, the output $X_{\text{out}}$ of multidirectional distance attention and output $Y_d$ of the partial fusion branch are adopted as inputs for the confidence score branch, allowing indirect information exchange to achieve unified prediction.

### E. Loss Function

The outputs of multiple detection heads are integrated for loss calculations during training to facilitate faster fitting of multiscale detection network. Training loss is mainly composed of three parts: classification loss, regression loss, and confidence loss. The calculation process of total loss is summarized as follows:

$$\begin{aligned} L = & \frac{\lambda_1}{N_{\text{pos}}} \sum L_{\text{cls}}(C, C^*) \\ & + \frac{\lambda_2}{N_{\text{pos}}} \sum L_{\text{reg}}(B, B^*) \\ & + \lambda_3 L_{\text{conf}}(P, P^*) \end{aligned} \tag{7}$$

where $N_{\text{pos}}$ and $L_{\text{cls}}$ represent the number of positive samples and classification loss, respectively, and $L_{\text{cls}}$ is calculated by adopting cross-entropy loss. Furthermore, $C$ refers to the predicted species and $C*$ denotes the real category. $L_{\text{reg}}$ represents the positioning loss and the generalized intersection over union (GIoU) loss is utilized to obtain it. $B$ represents the position of the prediction box, and $B*$ denotes the position of the real box. $L_{\text{conf}}$ stands for target confidence loss, and cross-entropy loss is also applied to generate it. Specifically, both classification and regression need the participation of positive samples, while all samples are required to participate in confidence calculation. And the hyperparameters $\lambda_1$, $\lambda_2$, and $\lambda_3$ are set to 1, 5, and 1, respectively.

## IV. Experiments

### A. Datasets

*1) DIOR [59] Dataset:* DIOR is the largest large-scale benchmark dataset currently utilized for object detection in optical RSIs. Targets have scale variations, extreme aspect ratios, high levels of interclass similarities, and intraclass diversity. Furthermore, the environment in the dataset is complicated, including pictures of various sceneries and weather conditions, which present huge challenges for evaluating the performance of detection models. The size of images in this dataset is 800 × 800, with a spatial resolution range of 0.5–30 m. There are totally 20 ground target categories in the dataset, and they are, respectively, airplane (AL), airport (AT), baseball field (BF), basketball court (BC), bridge (B), chimney (C), dam (D), expressway service area (ESA), expressway toll station (ETS), harbor (HB), golf course (GC), ground track field (GTF), overpass (O), ship (S), stadium (SD), storage tank (ST), tennis court (TC), train station (TS), vehicle (V), and windmill (WM). According to the instructions of the original paper, the entire dataset is divided into 5862 training images, 5863 validation pictures, and 11 738 test images.

*2) HRRSD [60] Dataset:* The HRRSD is a large-scale remote sensing dataset containing 21 761 satellite images from Google Earth and Baidu Map, with a spatial resolution ranging 0.15–12 m. There are 55 740 instances of targets in HRRSD, whose highlight is that the sample size is relatively balanced among the categories, with about 4000 samples in each category. HRRSD is composed of 13 classes and their species and proportions are as follows: ship (8.90%), bridge (8.20%), ground track field (6.70%), storage tank (7.90%), basketball court (6.95%), tennis court (7.71%), airplane (8.79%), baseball diamond (7.24%), harbor (6.70%), vehicle (8.53%), crossroad (7.22%), T junction (7.93%), and parking lot (8.90%). To confirm that our experimental results are more representative, HRRSD is divided into three parts, with 5401 images as a training set, 5417 images as a validation set, and 10 943 images as a test set, and each of the subsets maintains a relatively balanced number of images and annotated object instances.

*3) NWPU VHR-10 [61] Dataset:* It is an object detection dataset released by Northwestern Polytechnical University and is currently the most used in the field of remote sensing. In this dataset, 715 images with a spatial resolution of 0.5–2 m are captured from Google maps and the remaining 85 images with a spatial resolution of 0.08 m are acquired from sharpened infrared images. And the image sizes of this dataset are range from 533 × 597 to 1720 × 1028 pixels. This dataset consist of 150 negative samples and 650 samples with ten species. And their categories and proportions are as follows: airplane (19.43%), ship (7.75%), storage tank (16.81%, ST), base-ball diamond (10.01%, BD), tennis court (13.45%, TC), basketball court (4.08%, BC), ground track field (4.18%, GTF), harbor (5.57%), bridge (3.18%), and vehicle (15.35%). To demonstrate the effectiveness of our proposed method, we divide the dataset into training and test sets in a ratio of 0.75:0.25.

*4) RSOD [62] Datasets:* This is a dataset for object detection in RSI analysis published by Wuhan University in 2015. It contains a total of 2326 images, as well as four categories: airplanes, playgrounds, overpasses, and oil tanks. Their distribution in the dataset is as follows: 446 airplane images containing 4993 instances, 189 playground images containing 191 playgrounds, 176 overpass icons containing 180 overpasses, and 165 oil tank images containing 1586 oil tanks. The image size in the dataset varies from 512 × 512 to 1961 × 1193 pixels. In order to illustrate the effectiveness of our proposed method, we divide it into training set and test set according to the ratio of 0.6:0.4.

### B. Implementation Details

The backbone network ResNet50 is pretrained on the ImageNet dataset and then fine-tuned on RSIs. We utilize the

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Fig. 8. Some representative detection results of our detector on DIOR dataset.

mosaic data augmentation method and adjust the probability of random mosaic to 0.5. The stochastic gradient descent (SGD) optimizer is adopted and the momentum of optimizer is 0.9. The learning rate is initialized as 0.001, and it is adjusted with cosine decay to 0.0001. The training batch size is set to 6, and the size of inputs from four datasets is regulated into 640 × 640 pixels. Our experimental environment is the PyTorch framework under windows operating system, and all experiments are implemented on workstation with RTX4000 8-GB GPU and 32-GB RAM.

### C. Evaluation Metrics

To comprehensively assess the performance of our proposed object detection algorithm, the widely used average precision (AP) is introduced as an evaluation metric. AP is achieved by calculating the area under the precision-recall curve. Precision and recall are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{9}$$

where TP, FP, and FN denote the true positives, false positives, and false negatives, respectively. The intersection over union (IoU) is the overlap rate between the predicted and true boxes, and it is an important parameter in object detection. To be specific, we regard the detection result as TP if IoU between the predicted box and the ground-truth box is bigger than 0.5. Otherwise, it is assigned to be an FP. In order to measure the detection performance of multiple targets, we average the APs of multiple species to obtain mean average precision (mAP), and the whole calculation process is concluded as follows:

$$\text{AP} = \int_0^1 P(R) dR \tag{10}$$

$$\text{mAP} = \frac{1}{n} \sum_{i=1}^{n} \text{AP}_i \tag{11}$$

where $P$ represents precision and $R$ refers recall. Besides, $n$ denotes the number of categories, and $i$ indicates the category labels. $\text{AP}_i$ represents the AP of the $i$th category.

### D. Comparison With the State-of-the-Arts

In order to demonstrate our detector has excellent performance and better generalization ability in the field of remote sensing, comparative experiments are implemented on three public datasets. We also compare the prominent performance of our approach with various state-of-the-art networks, including Faster R-CNN [7], Cascade R-CNN [56], Retinanet [13], YOLOv4 [10], YOLOv5 [11],YOLOv7 [63], YOLOv8 [64], FCOS [14], CornerNet [16], CenterNet [15], SRAF-Net [43], MSFC-Net [57], ASSD [51], FSoD-Net [35], CANet [65], EGAT [26], SSA-LSTM [26], ABNet [38], DETR [68], DIAG-TR [69], RAST-YOLO [70], and more. The algorithms comprise not just one-stage, two-stage, and anchor-free networks, but also state-of-the-art detection networks for natural images, RSI detection networks, and transformer-based detection networks. More details are provided as follows.

*1) Evaluation on DIOR:* DIOR dataset comprises targets of various scales and complex scenes, which can effectively verify the superiority and generalization ability of the algorithms involved in overcoming these two obstacles. According to the data presented in Table I, the transformer-based algorithms obtain good detection accuracy, possibly due to their more comprehensive global modeling ability. At the same time, our detection algorithm achieves the highest mAP of 72.9% compared to other advanced networks, which may be attributed to our algorithm's ability to more effectively focus on target areas in complex environments. Results in Table I also exhibit that our network has superior performance of seven categories. To be specific, our SCTB detection head pays more attention to spatial distance and multichannel aggregation information, and also makes the network better align in positioning and classification tasks. This character allows our detector to achieve 47.1%, 61.4%, and 64.9% AP on the categories with

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TABLE I
COMPARISON WITH STATE-OF-THE-ART ON DIOR. AND THE BEST RESULTS ARE MARKED IN BOLD

| methods | mAP | AL | AT | BF | BC | B | C | D | ESA | ETS | GC | GTF | HB | O | S | SD | ST | TC | TS | V | WM |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Faster R-CNN | 66.9 | 81.1 | 76.9 | 89.8 | 80.5 | 45.1 | 81.7 | 63.4 | 80.9 | 63.2 | 73.3 | 78.3 | 43.3 | 58.4 | 68.5 | **89.1** | 59.0 | 81.1 | 60.2 | 47.8 | 79.3 |
| Cascade R-CNN | 67.4 | 81.2 | 81.4 | 90.1 | 81.1 | 46.9 | 81.5 | **68.1** | 84.2 | 65.3 | 74.5 | 81.6 | 37.8 | 60.4 | 68.9 | 88.8 | 60.5 | 81.0 | 57.5 | 47.7 | 80.7 |
| RetinaNet | 66.1 | 53.3 | 77.0 | 69.3 | 85.0 | 44.1 | 73.2 | 62.4 | 78.6 | 62.8 | 78.6 | 76.6 | 49.9 | 59.6 | 71.1 | 68.4 | 45.8 | 81.3 | 55.2 | 44.4 | 85.5 |
| YOLO-v4 | 66.4 | 79.3 | 71.2 | 70.2 | 88.7 | 39.2 | 77.1 | 48.6 | 58.2 | 58.3 | 75.5 | 71.2 | 53.8 | 57.7 | 90.1 | 51.0 | 77.5 | 83.3 | 46.7 | 55.1 | 75.1 |
| YOLO-v5 | 68.6 | 87.3 | 61.7 | 73.8 | 90.0 | 42.6 | 77.5 | 55.2 | 63.8 | 63.2 | 66.9 | 78.0 | 58.2 | 58.1 | 87.8 | 54.3 | **79.3** | 89.7 | 50.2 | 54.0 | 79.6 |
| FCOS | 61.2 | 73.5 | 68.0 | 69.9 | 85.1 | 34.7 | 73.6 | 49.3 | 52.1 | 47.6 | 67.2 | 68.7 | 46.3 | 51.1 | 72.2 | 59.8 | 64.6 | 81.2 | 42.7 | 42.2 | 74.8 |
| CornerNet | 65.9 | 58.8 | **84.2** | 72.0 | 80.8 | 46.4 | 75.3 | 64.3 | 81.6 | **76.3** | 79.5 | 79.5 | 26.1 | 60.6 | 37.6 | 70.7 | 45.2 | 84.0 | 57.1 | 43.0 | 75.9 |
| CenterNet | 63.9 | 73.6 | 58.0 | 69.7 | 88.5 | 36.2 | 76.9 | 47.9 | 52.7 | 53.9 | 60.5 | 62.6 | 45.7 | 52.6 | 88.2 | 63.7 | 76.2 | 83.7 | 51.3 | 54.4 | 79.5 |
| SRAF-Net | 69.7 | 88.4 | 76.5 | **92.6** | 87.9 | 35.8 | **83.8** | 58.6 | 86.8 | 66.8 | 76.4 | **82.8** | 16.2 | 58.0 | 59.4 | 80.9 | 55.6 | 90.6 | 52.0 | 53.2 | **91.0** |
| MSFC-Net | 70.1 | 85.8 | 76.2 | 74.4 | 91.1 | 44.2 | 78.1 | 55.5 | 60.9 | 59.5 | 76.9 | 73.7 | 49.6 | 57.2 | 89.6 | 69.2 | 76.5 | 86.7 | 51.8 | 55.2 | 84.3 |
| DETR | 68.6 | 39.6 | 74.0 | 65.2 | 80.7 | 26.5 | 75.2 | 66.8 | 70.5 | 52.5 | 74.2 | 62.1 | 27.4 | 47.0 | 8.5 | 46.0 | 14.5 | 64.4 | 54.3 | 14.4 | 55.6 |
| DIAG-TR | 72.0 | 68.4 | 72.2 | 71.9 | 85.1 | 29.8 | 76.8 | 60.2 | 70.3 | 73.2 | 75.1 | 35.5 | 35.5 | 50.8 | 41.6 | 71.8 | 32.9 | 81.0 | 53.1 | 33.8 | 78.6 |
| RAST-YOLO | 69.8 | 84.3 | 76.4 | 78.7 | 85.9 | 40.2 | 76.8 | 50.2 | 62.6 | 56.5 | 77.1 | 73.7 | 61.1 | 56.6 | **91.1** | 74.3 | 77.9 | 89.3 | 53.3 | 54.0 | 76.2 |
| ASSD | 71.1 | 85.6 | 82.4 | 75.8 | 89.5 | 40.7 | 77.6 | 64.7 | 67.1 | 61.7 | **80.8** | 78.6 | 62.0 | 58.0 | 84.9 | 76.7 | 65.3 | 87.9 | **62.4** | 44.5 | 76.3 |
| FSoD-Net | 71.8 | 88.9 | 66.9 | 86.8 | 90.2 | 45.5 | 79.6 | 48.2 | **86.9** | 75.5 | 67.0 | 77.3 | 53.6 | 59.7 | 78.3 | 69.9 | 75.0 | **91.4** | 52.3 | 52.0 | 90.6 |
| Ours | **72.9** | **92.2** | 77.7 | 75.0 | **91.3** | **47.1** | 78.6 | 53.6 | 67.1 | 66.2 | 78.5 | 76.3 | **64.9** | **61.4** | 90.5 | 72.2 | 75.4 | 90.7 | 62.1 | **55.2** | 83.2 |

TABLE II
COMPARISON WITH STATE-OF-THE-ART ON NWPU VHR-10. AND THE BEST RESULTS ARE MARKED IN BOLD

| Methods | mAP | airplane | ship | ST | BD | TC | BC | GTF | harbor | bridge | vehicle |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Faster R-CNN | 88.30 | 99.70 | 85.58 | 99.27 | 95.93 | 88.22 | 92.08 | 99.73 | 92.11 | 43.37 | 86.60 |
| Cascade R-CNN | 92.19 | 99.54 | 88.53 | 95.98 | 94.46 | 94.02 | 88.21 | 97.16 | 91.45 | 82.30 | 90.25 |
| YOLO-v4 | 90.77 | 99.40 | 92.63 | **99.81** | 95.27 | 98.30 | 75.22 | 94.17 | 93.12 | 71.95 | 87.80 |
| YOLO-v5 | 92.3 | 99.50 | 94.50 | 76.40 | 99.00 | 90.50 | 97.20 | 99.50 | **97.40** | 76.00 | 92.70 |
| YOLO-v7 | 93.31 | 99.74 | 92.71 | 97.69 | 97.87 | 93.97 | 91.83 | 95.63 | 89.51 | 83.75 | 90.47 |
| YOLO-v8 | 93.95 | 99.99 | 94.37 | 96.78 | 98.12 | 94.36 | 90.77 | 97.25 | 91.33 | 85.7 | 91.02 |
| FCOS | 92.14 | 99.99 | 85.21 | 96.94 | 97.75 | 95.80 | 80.34 | 99.67 | 95.04 | 81.82 | 88.92 |
| CANet [65] | 93.33 | 99.99 | 85.99 | 99.27 | 97.28 | 97.80 | 84.77 | 98.38 | 90.38 | 89.16 | 90.25 |
| EGAT [26] | 92.01 | 97.32 | **96.72** | 97.16 | 96.51 | 86.64 | 94.46 | 94.18 | 86.18 | 80.14 | 90.79 |
| ABNet [38] | 94.21 | **100** | 92.58 | 97.77 | 97.76 | **99.26** | 95.98 | **99.86** | 94.26 | 69.04 | **95.62** |
| Ours | **95.29** | 99.86 | 93.40 | 95.95 | **99.59** | 96.43 | **97.65** | 98.18 | 94.03 | **89.60** | 88.38 |

complex aspect ratios such as bridge, overpass, and harbor. In addition as shown in Fig. 8, our network has also exhibited excellent performance in some cluster categories, such as Airplane of 92.2% AP, Vehicle of 55.2% AP, Basketball Court of 91.3% AP, and Ship of 90.5% AP. This is probably due to HRFPN, which effectively highlights the characteristics of cluster targets and eliminate the influence of complicated environments with a global view.

*2) Evaluation on NWPU VHR-10:* As illustrated in Fig. 9, the NWPU VHR-10 (NWPU) dataset exhibits significant spectral variations and a certain degree of image distortion due to imaging height and angle. As a result, conducting comparative experiments on this dataset can yield more representative and convincing results. Table II demonstrates that our proposed algorithm achieves the highest comprehensive detection performance with a 95.29% mAP, compared with other state-of-the-art algorithms for natural image detection and RSI detection, and presents improvements in mAP ranging from 7% to 1.08%. Furthermore, in terms of baseball diamond, basketball court, and bridge, our detector acquires wonderful detection performance among all methods. Although ABnet receives the highest AP in four classes, and our algorithm has a more balanced and outstanding performance across multiple categories. The data in Table II show that our network is not only competitive for complex distorted targets, but also better adaptive ability for multiscale targets.

In addition, the presence of degradation, noise, and spectral variabilities can significantly affect the quality and accuracy of the image analysis tasks. Therefore, it is necessary to verify whether our detection algorithm can have good robustness and generalization ability when encountering multiple environmental changes. By separately adding Gaussian white noise, spectral variations, and fog blur to the NWPU VHR-10 dataset, three datasets with complex interference are generated. The HRFPN output heatmaps and detection results for different scenarios are illustrated in Fig. 10. It is apparent that the

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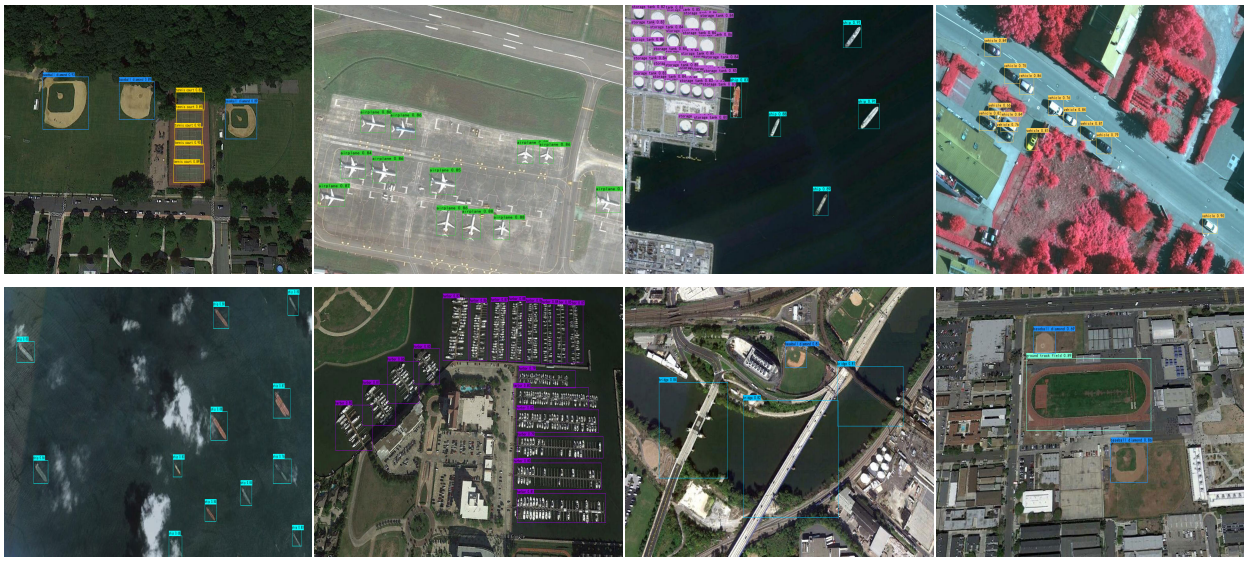


Fig. 9. Visualization of results on NWPU VHR-10.

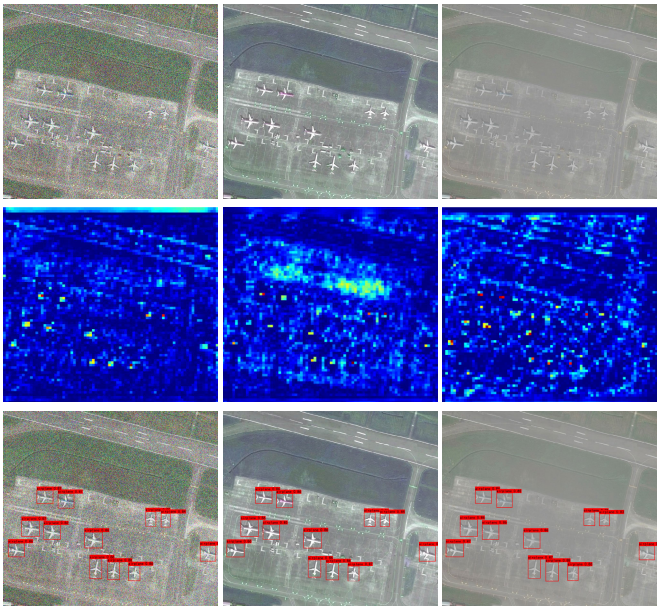



Fig. 10. Visualizations of detection results in four different environments. (a)–(c) Images with added noise, spectral changes, and fog, respectively.

detection network can accurately detect the objects in different scenes, and by comparing the heat maps, it is observed that the information focus of the detection network does not change significantly among the other three environments, and is able to concentrate effectively on the target regions. In addition, we also found that compared to the original data, our detection algorithm has a slight decrease in mAP in these scenarios. The dataset with added Gaussian white noise has the most significant drop, with a decrease of about 2.23% in mAP, while the other two complex environments only caused a decrease of less than 1.3% in mAP. By comparing the detection accuracy of the baseline methods on three complex environmental datasets, all of them showed a decrease, with an average decrease of 2.37% mAP. This clearly indicates that the decrease in our method is smaller, which can also indirectly demonstrate the better generalization ability of our detection algorithm.

*3) Evaluation on HRRSD:* Reasonable comparisons are implemented between our detector and various excellent detectors, including one-stage, two-stage, and networks specifically designed for the remote sensing, on HRRSD dataset. As illustrated in Table III, our detector not only obtains the highest 92.96% mAP but also achieves superior performance in eight categories. The results speak volumes that our network is more sensitive to multiscale targets and can effectively mine the relationship between global pixels to distinguish targets. Compared with the two-stage networks, our network has higher training efficiency and better target anchoring ability than anchor-based networks. In addition, the visualization results in Fig. 11 effectively show that large scale and small scale targets, as well as large horizontal and horizontal targets can be accurately detected.

### *E. Ablation Experiments*

In order to comprehensively evaluate the superiority and effectiveness of our proposed method, ablation experiments are conducted on the NWPU and RSOD datasets. The mAP, model parameter size, floating-point operations (FLOPs), and inference time during the experimental process are adopted as the main evaluation criteria. To be fair, all of experiments adopt the same hyperparameters and training strategy, and the loss function employed is unchanged. First, a detector with ResNet50 feature extraction backbone, FPN, and conventional decoupling head is regarded as our baseline network. The information in Table IV points out that baseline network has 38.426M model parameters, 58.435G FLOPs, and an inference time of 0.625 s. Moreover, it achieves 92.24% and 91.93% mAP on NWPU and RSOD datasets, respectively.

Subsequently, our proposed modules are embedded in baseline network and their superiority is systematically analyzed based on detection performance and model complexity. DSGC is designed to obtain valid information about the global

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Fig. 11. Visualization of detection results on HRRSD dataset.

TABLE III

COMPARISON WITH STATE-OF-THE-ART ON HRRSD. AND THE BEST RESULTS ARE MARKED IN BOLD

| Methods | mAP | AL | S | ST | BD | TC | BC | GTF | HB | B | V | CR | TJ | PK |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Faster R-CNN | 81.5 | 90.8 | 88.5 | 88.7 | 86.9 | 80.7 | 47.9 | 90.6 | 89.4 | 85.5 | 84.0 | 88.6 | 75.1 | 63.3 |
| Cascade R-CNN | 89.1 | 98.00 | 91.40 | 95.60 | 91.30 | 92.60 | 72.20 | 98.20 | 93.90 | 89.60 | 95.50 | 92.30 | 79.70 | 68.40 |
| MGCN [66] | 86.27 | 93.27 | 88.96 | 92.10 | 91.73 | 83.98 | 55.79 | 93.46 | 92.96 | 89.79 | 89.85 | 94.95 | 79.87 | 74.69 |
| YOLOv4 | 90.90 | 98.70 | 91.90 | 96.30 | 91.50 | 93.00 | 81.20 | 95.40 | 93.40 | 90.70 | 93.80 | 95.10 | 84.90 | 75.80 |
| SSA-LSTM [26] | 91.30 | 95.67 | 93.92 | 93.45 | **94.52** | 85.89 | **86.63** | 95.72 | 94.36 | 91.23 | 89.76 | **95.86** | 83.71 | **86.22** |
| SB-MSN [67] | 92.60 | 99.30 | **94.50** | 97.80 | 93.90 | 96.00 | 80.20 | 99.00 | **96.00** | 92.70 | 96.60 | 95.30 | 86.40 | 76.50 |
| ours | **92.96** | **99.90** | 93.34 | **97.96** | 93.69 | **96.07** | 83.99 | **99.22** | 95.21 | **95.14** | **96.93** | **95.86** | **87.64** | 73.46 |

TABLE IV

ABLATION STUDIES ON NWPU AND RSOD DATASETS. THE "PARAM" REFERS TO THE SIZE OF THE MODEL'S PARAMETERS, WHILE "INFER.TIME" REPRESENTS THE INFERENCE TIME

| Methods | DSGC | HRFPN | SCTB | mAP on NWPU (%) | Gain | mAP on RSOD(%) | Gain | Param(M) | FLOPs(G) | Infer.time(s) |
|---|---|---|---|---|---|---|---|---|---|---|
| Baseline | X | X | X | 92.24 | - | 91.93 | - | 38.426 | 58.435 | 0.0625 |
| V1 | √ | X | X | 92.95 | +0.71 | 92.87 | +0.94 | 40.005 | 59.069 | 0.0632 |
| V2 | X | √ | X | 93.04 | +0.80 | 92.65 | +0.72 | 40.261 | 60.129 | 0.0639 |
| V3 | X | X | √ | 93.35 | +1.11 | 92.98 | +1.07 | 38.159 | 57.706 | 0.0647 |
| V4 | √ | √ | X | 93.72 | +1.48 | 93.53 | +1.60 | 41.840 | 60.763 | 0.0656 |
| V5 | √ | X | √ | 94.21 | +1.97 | 93.45 | +1.52 | 39.738 | 58.340 | 0.0666 |
| V6 | X | √ | √ | 94.07 | +1.83 | 93.62 | +1.70 | 39.994 | 59.400 | 0.0673 |
| TMAFNet | √ | √ | √ | 95.23 | +2.99 | 94.10 | +2.27 | 41.573 | 60.034 | 0.0681 |

context, which may be instrumental to better understand the relationship between the environments and targets. As shown from Table IV, by inserting the DSGC module to the baseline network, the performance of the network is promoted to 92.95% mAP on NWPU and 92.87% mAP on RSOD. At this point, the network's parameter count and FLOPs are 40.005M and 58.069G, respectively, which represent a marginal increase of only 1.579M and 0.634G over the baseline network. Additionally, Table V compares DSGC with other relevant local context extraction methods, such as SE [37], CBAM [22], and global context extraction methods such as W-MSA [25] and self-attention [24]. It is evident that DSGC module, as a global context extraction method, introduces comparable parameter and computational complexity to local context methods. More importantly, the DSGC outperforms global context methods in terms of the highest detection performance improvement while significantly reducing the parameters and computational complexity. HRFPN is constructed to enhance object feature extraction through multiscale feature fusion while filtering out interference from the background. Replacing FPN in the baseline network with HRFPN results in a performance improvement of 0.80% and 0.72% mAP on the NWPU and RSOD datasets, respectively. As shown in Tables IV and VI, the increase in the network's model parameters and

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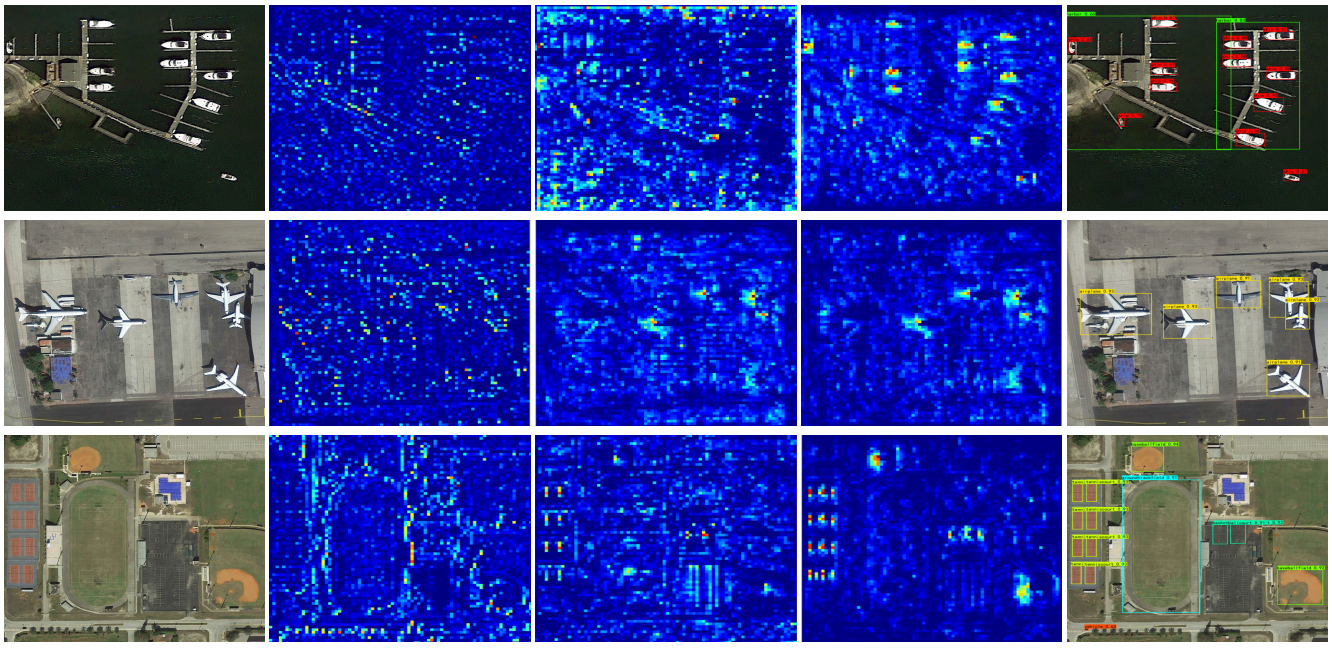



Fig. 12. Visualizing the feature extraction effects. (a) Input images for detection. (b)–(d) Heat maps generated from the input feature layers $C_i$, intermediate feature layers $P_i$, and output feature layers $P_i'$ of HRFPN, respectively. (e) Detection result images.

TABLE V
COMPARISON WITH SE, CBAM, W-MSA, AND SELF-ATTENTION ON NWPU VHR-10 DATASET

| Methods | $mAP_{50}(\%)$ | Parameters(M) | FLOPs(G) |
|---|---|---|---|
| Baseline | 92.24 | 38.426 | 58.435 |
| SE | 92.73(+0.49) | 39.744 | 58.436 |
| CBAM | 92.66(+0.42) | 39.744 | 58.436 |
| W-MSA | 92.88(+0.64) | 41.975 | 59.872 |
| Self-Attention | 92.84(+0.60) | 42.628 | 60.319 |
| DSGC | 92.95(+0.71) | 40.005 | 59.069 |

TABLE VI
COMPARISON WITH PANET, SMSL, FRP, AND RFP ON NWPU VHR-10 DATASET

| Methods | $mAP_{50}(\%)$ | Parameters(M) | FLOPs(G) |
|---|---|---|---|
| Baseline | 92.24 | 38.426 | 58.435 |
| PANet [19] | 92.67(+0.43) | 41.565 | 62.573 |
| SMSL [40] | 92.93(+0.69) | 42.157 | 60.975 |
| FRP [46] | 92.89(+0.65) | 42.557 | 61.572 |
| RFP [41] | 92.99(+0.75) | 42.137 | 63.157 |
| HRFPN | 93.04(+0.80) | 40.261 | 60.129 |

computational complexity is only 1.835M and 1.649G, respectively. In comparison to other multiscale fusion methods, such as PANet [19], selective multiscale learning (SMSL) [40], FRP [46], and RFP [41], HRFPN not only has the least number of parameters and computational complexity compared to advanced multiscale fusion networks, but also achieves the highest improvement in detection performance. This experiment further demonstrates that HRFPN has excellent performance compared to advanced multiscale fusion networks. As shown in Fig. 12, heat maps are generated to visualize the feature outputs of $C_i$, $P_i$, and $P_i'$ in Fig. 3. It is evident that the output of HRFPN is not only more focused on the target region compared to the input $C_i$ and the multiscale fused feature $P_i$, but it also effectively eliminates a significant amount of background interference. This observation provides further evidence of HRFPN's effectiveness in enhancing target feature discriminability and reducing the impact of background information, resulting in improved detection performance.

Regarding the SCTB detection head, the original intention is to address the phenomenon of misalignment of classification and positioning tasks caused by complex environments or large aspect ratios, while simultaneously providing each subtask with the necessary information. Upon replacing the decoupled detection head with the proposed SCTB detection head, the detection algorithm exhibits a decrease in both parameter count and computational complexity, with respective detection accuracies of 93.35% and 92.98% mAP on the NWPU and RSOD datasets, demonstrating a performance increase of 1.11% and 1.07% mAP over the baseline network. Notably, as illustrated in Fig. 13, we present the heatmaps of the two subtask branches of the decoupled detection head, as well as the heatmaps of the two branches of the SCTB detection head. It is evident that our proposed algorithm maintains better consistency in both spatial and focused information. Table VII demonstrates that although the coupled detection head has the least number of parameters and FLOPs, it incurs a loss of 0.85% mAP. It is noteworthy that SCTB achieves the highest detection performance improvement while requiring the least amount of computational cost when compared to the decoupled and T-head approaches. Overall, SCTB shows competitive performance compared to some of the excellent detection heads [47].

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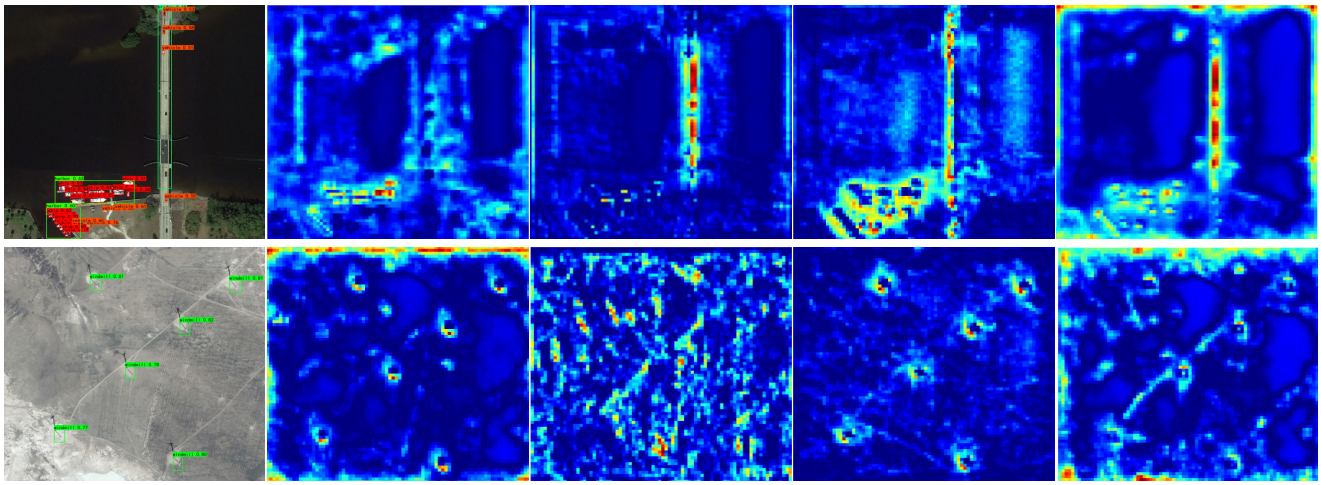



Fig. 13. Visualizing the feature extraction effects. (a) Original image and visualization of detection result. (b) and (c) Heat maps of decoupled classification and location branches. (d) and (e) Heat maps of our classification and location.

TABLE VII
COMPARISON WITH DECOUPLED AND COUPLED HEADS AND T-HEAD ON NWPU VHR-10 DATASET

| Methods | $mAP_{50}(\%)$ | Parameters(M) | FLOPs(G) |
|---|---|---|---|
| Baseline | 92.24 | 38.426 | 58.435 |
| Coupled [10] | 91.32(-0.85) | 36.927 | 57.172 |
| T-head [47] | 92.84(+0.97) | 38.549 | 57.737 |
| SCTB | 93.35(+1.11) | 38.159 | 57.706 |

The combination of these three modules results in a cooperative effect that is observable in Table IV, leading to an improved performance of the detection network. Ultimately, after embedding these three modules into the baseline network, our detection network achieves a detection accuracy of 95.23% and 94.1% mAP on NWPU and RSOD datasets, respectively. Compared to the baseline network, our approach increases the model parameters by 3.147M, FLOPs by 1.559G, and inference time by 0.0056 s. Overall, our algorithm produces a nearly three-point improvement in mAP with a 2.7% increase in computational cost.

## V. CONCLUSION

A TMAFNet is explored for the detection of RSIs, which has prominent performance for dealing with complicated environments and targets with vaster variances. DSGC and HRFPN are easier to eliminate interference from background and focus on targets information in a novel way. SCTB not only supplies multichannel aggregated expressions and multi-directional distance information, but also constructs a balance between classification and positioning. Abundant experiments illustrate that our detector is competitive with some state-of-the-art algorithms on several public remote sensing datasets. However, our detector has room for improvement in accurately detecting small objects. In the future, we will focus on improving the detection of small and dense objects in complex environments. Additionally, we will explore the possibility of implementing lightweight network architectures without compromising detection accuracy.

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**Tao Gao** (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China, in 2002, 2010, and 2016, respectively.

He is currently a Professor with the School of Information Engineering, Chang'an University, Xi'an. His research interests include image processing and computer vision.



**Zixiang Liu** received the B.Sc. degree from the School of Information Engineering, Chang'an University, Xi'an, China, in 2021, where he is currently pursuing the M.Sc. degree.

His research interests include computer vision and intelligent transportation.



**Jing Zhang** currently holds a post-doctoral position with the School of Computing, The Australian National University, Canberra, ACT, Australia. Her main research interests include saliency detection, weakly supervised learning, and generative model.

Dr. Zhang received the Best Student Paper Prize at DICTA 2017, the Best Deep/Machine Learning Paper Prize at APSIPA ASC 2017, and the Best Paper Award Nominee at IEEE CVPR 2020.



**Guiping Wu** received the B.Sc. degree from the School of Information Engineering, Chang'an University, Xi'an, China, in 2021, where she is currently pursuing the M.Sc. degree.

Her research interests include computer vision and intelligent transportation.



**Ting Chen** received the Ph.D. degree in information and communication engineering from Xidian University, Xi'an, China, in 2011.

She is currently an Assistant Professor with the School of Information Engineering, Chang'an University, Xi'an. Her research interests include image processing and wireless network.

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